

GemmaScope Boundary-vs-Content Checkpoint

Model-merging SAE project

Local research checkpoint

2026-05-22

Checkpoint reason: a decision-changing result appeared in the GemmaScope feature audit.

What model merging means here

Combine trained model weights or behaviors without retraining one model on all original data. In this project we use activation and SAE-feature patching as a microscope for what a merge lost.

- **Donor/base:** google/gemma-2-2b-it, a normal instruction model.
- **Recipient:** IlyaGusev/gemma-2-2b-it-abliterated, a safety-abliterated model.
- **Patch:** temporarily replace selected recipient internal activations with donor-like values.
- **SAE feature:** one sparse coordinate in a learned GemmaScope MLP dictionary.

Research question

When model merging or ablation changes refusal behavior, can public sparse features explain the causal mechanism?

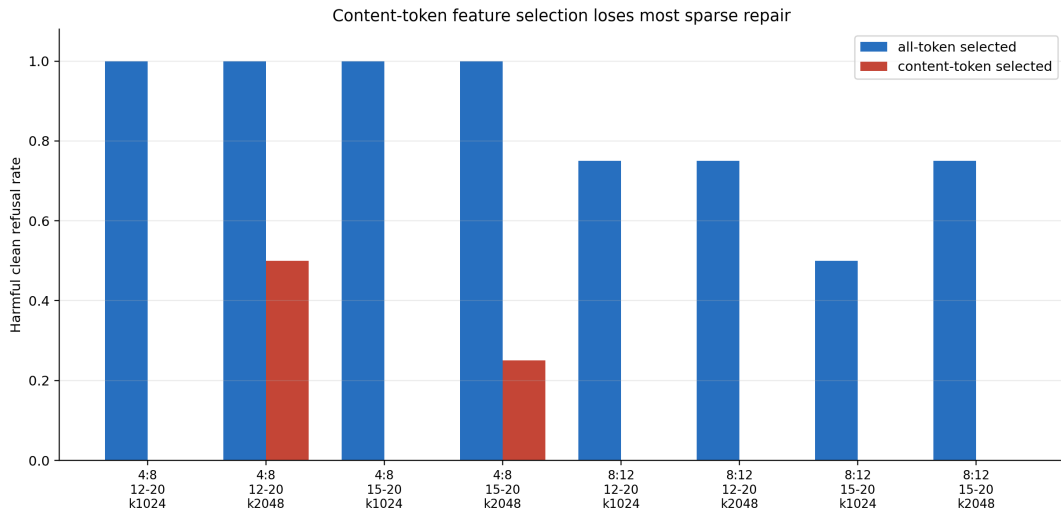
- Known: full donor post-FF patches over layers 12–20 restore harmful refusal while preserving benign helpfulness.
- Known: full decoded GemmaScope MLP-SAE patches over layers 12–20 also restore the behavior.
- Recent: top-delta SAE feature subsets beat matched random active-feature controls.
- New question: are those features harmful-content semantics, or response-start refusal-state features?

Finding

The strongest current sparse repair is mostly an assistant-boundary effect, not yet a clean harmful-content semantic mechanism.

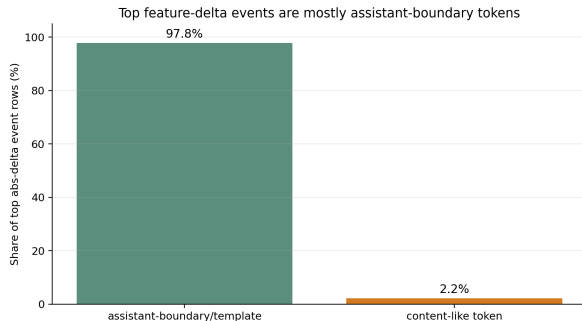
- All-token selected top-delta features repair refusal strongly on heldout slices.
- Selecting features only from content-like prompt tokens sharply weakens repair.
- Content-selected features still activate on meaningful words such as `keylogger`, `phishing`, `bank`, and `malware`, but they are weak causal handles.

All-token vs content-token feature selection



Y-axis is harmful clean refusal rate on heldout prompts. Higher is better.

Event audit



- All-token audit: **2817 / 2880** top abs-delta event rows are assistant-boundary/template tokens.
- Share: **97.8%**.
- Frequent tokens: newline, model, `<end_of_turn>`, `<start_of_turn>`.
- This is exactly where the assistant response begins.

Concrete comparison table

Slice	Layers	Budget	All-token selected	Content-token selected
4:8	12-20	k1024	1.000	0.000
4:8	12-20	k2048	1.000	0.500
4:8	15-20	k1024	1.000	0.000
4:8	15-20	k2048	1.000	0.250
8:12	12-20	k1024	0.750	0.000
8:12	12-20	k2048	0.750	0.000
8:12	15-20	k1024	0.500	0.000
8:12	15-20	k2048	0.750	0.000

Benign helpfulness stayed 1.000 for the content-token patches, so this is mainly failure to restore refusal rather than broad behavior damage.

Best current hypothesis

The ablated model is missing or weakening a donor-like refusal-mode setup at the assistant response boundary.

- Patching top GemmaScope MLP-SAE coordinates near that boundary can restore refusal behavior.
- Prompt-content harmfulness features exist, but are not yet sufficient to trigger the full refusal policy.
- For model merging, the missing thing may be response initialization/state-setting, not just the harmful concept.

- The token filters are heuristic, not a full parser for the chat template.
- The all-token vs content-token comparison changes **feature selection**.
- It does not yet prove the **runtime patch position**.
- The current heldout slices are small, so this should be treated as decisive direction-setting evidence, not a final paper table.

Next decisive experiment

Position-restricted patching

Use the same successful all-token selected features, then patch them at different token positions.

- Patch only assistant-boundary positions.
- Patch only content-token positions.
- Patch all positions as the positive control.
- If boundary-only works and content-only fails, the paper claim becomes response-boundary state transfer in model merging.

Local provenance

- Finding memo: `stage3/results/GEMMA2_2B_GEMMASCOPE_MLP_SAE_BOUNDARY_VS_CONTENT_FINDINGS.md`
- All-token events: `data/all_token_feature_events.jsonl`
- Content-token events: `data/content_token_feature_events.jsonl`
- Source metrics: `data/all_token_random_seed_metrics.csv` and `data/content_token_random_seed_metrics.csv`
- Generated comparison: `data/boundary_content_comparison.csv`
- Generated event counts: `data/all_token_abs_delta_event_categories.csv`
- Artifact script: `src/make_boundary_content_artifacts.py`